Fiscal Year:	FY 2015	**Task Last Updated:**	FY 03/04/2015
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Project Title:	Evaluation of the Medical Food, Enterade-r, as a Radiation Countermeasure for Enteric Dysfunction – SMARTCAP-Advanced		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI-RFA-SMARTCAP
Start Date:	04/01/2014	**End Date:**	11/30/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-RE00007		
Performance Goal No.:			
Performance Goal Text:			
	Original project aims/objectives: Radiation-induced gastrointestinal (GI) toxicity and the corresponding symptoms of nausea and vomiting are of particular concern on a spaceflight of extended duration. However, research on emesis and antiemetics has been limited, partly because common laboratory animals, e.g., rats and mice, do not vomit. Pica, the eating of nonnutritive substances, is an illness-response behavior in rodents that is indicative of emesis. Our work has focused on pica models, ion flux, and mineral loss studies to help us better understand the effect of Enterade on GI responses to low-dose gamma- and proton radiation. Original Aims: #1: Determine the effect of Enterade on changes in pica behavior in mice exposed to low-dose gamma radiation. #2: Determine the effect of Enterade on changes on pica behavior in mice exposed to low-dose proton radiation. Subsequent Aims: #1: Determine the chloride secretory and sodium absorptive patterns in the ileal tissues of gamma and		

proton irradiated mice (0, 0.5, 1, and 1.5 Gy). #2: Determine the intracellular calcium and cAMP levels in the ileal tissues of gamma and proton irradiated mice (0, 0.5, 1, and 1.5 Gy). #3. Use in vivo perfusion models and Ussing chambers to determine changes in the absorptive patterns and function of transport proteins for specific minerals in the presence and absence of gamma or proton radiation.

Key findings: In irradiated mice, Enterade treatment resulted in decreased kaolin intake (pica), improved fluid intake, and improved body weight with effects comparable to those of ondansetron, as well as improved radiological clearance and lung function. Proton and gamma radiation both caused anion secretion, but differed in induction of intracellular Ca2+- and cAMP-mediated chloride secretion. In preliminary studies, we observed radiation dose-dependent stool loss of zinc, magnesium, copper, and phosphorus with proton radiation, and Enterade reduced proton radiation-induced stool loss of zinc.

Impact of key findings on hypotheses, technology requirements, objectives, and specific aims of the original proposal: The effects of low-dose gamma- and proton radiation on pica behavior provided an effective means of evaluating anti-emetics. Gamma radiation-induced anion secretion was inhibited by calcium-activated chloride channel (CACC) blockers, whereas proton radiation-induced anion secretion was inhibited by blockers of both cAMP-activated cystic fibrosis transmembrane conductance regulator (CFTR) and CACC. The effects of Enterade observed with irradiated mice suggest that this treatment could help astronauts to overcome the nausea and vomiting that are common during space missions. Moreover, our studies show that proton radiation may result in sustained electrolyte and mineral loss through clinical or subclinical diarrhea, a mechanism that could explain the mineral loss associated with an extended stay in space.

Task Description:

Proposed research plan for the coming year: Our work indicates that Enterade – which is already commercially available online – has immediate potential for use in overcoming the nausea and vomiting that astronauts experience during space missions. Gamma radiation results in intracellular Ca2+-activated Cl- secretion, whereas proton radiation results in both Ca2+- and cAMP-dependent Cl- secretion. cAMP-dependent Cl- secretion is persistent, resulting in chronic-persistent and/or subclinical diarrhea, whereas Ca2+-activated Cl- secretion is transient, resulting in acute diarrhea. Previous studies have shown that mineral loss occurs with both acute and chronic diarrhea. We therefore hypothesize that proton radiation-induced diarrhea may result in mineral loss, potentially helping to explain the increase in bone demineralization associated with extended stays in space. In addition, certain minerals (e.g., zinc) have very low stores in the body, and the decreased mineral absorption associated with diarrhea could lead to rapid depletion of stores. Further studies are needed to determine the pathophysiological mechanism responsible for mineral loss. To better understand the mineral loss in stool and to help us to transition our work to a flight project, we will perform the following studies:

Aim#1A: Use in vivo perfusion models to determine changes in the absorptive patterns of specific minerals from the gut in the presence and absence of gamma or proton radiation and Enterade. To examine absorption patterns, in vivo perfusion models of the gut will be used in conjunction with various mineral isotopes in the presence or absence of gamma- or proton radiation, and with or without Enterade treatment. Mineral isotope levels in the lumen perfusate and plasma will be assessed. Our preliminary results suggest that proton radiation is associated with pronounced mineral loss in the stool, and that Enterade can mitigate this loss.

Aim #1B: Use in vitro Ussing chamber studies to determine changes in mineral transporters in the presence and absence of gamma or proton radiation. To examine mineral transport, Ussing chamber studies using specific mineral isotopes will be performed using from non-, proton-, or gamma-irradiated ileal tissues. Specific activity of the transporters will be determined in the presence of the substrate with or without mineral transport inhibitors.

Aim#2: Develop a biomarker to assess clinical or subclinical alterations in GI function secondary to low-dose irradiation. Intestinal tissues from irradiated mice show radiation dose-dependent increases in the Cl- channel anoctamin 1 (ANO1), a key transporter mediating radiation-induced diarrhea. We did MS/MS analysis and validated the identity of the protein. Since ANO1 protein levels on erythrocyte and leukocyte membranes also increase with radiation exposure, ANO1 expression in peripheral blood cells could be used as a surrogate marker for predicting Cl- secretory defects secondary to irradiation.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Approximately 400,000 patients in the US undergo abdominopelvic radiotherapy each year for cancer. These patients often suffer from acute radiation enteritis; however, current treatments are mostly based on symptomatology, including antidiarrheals to prevent fluid loss, opioids to relieve stomach and rectal pain, glucose-based electrolyte solutions to correct dehydration, and ondansetron for nausea and vomiting. Beyond the everyday clinical situation, in the case of a mass-casualty nuclear event, there is a need for an orally administered, CONOPS-friendly agent that is effective across various doses and schedules to overcome radiation-induced gastrointestinal (GI) toxicity. To meet the needs of cancer patients and homeland security, as well as astronauts experiencing radiation-induced nausea and vomiting, our team developed Enterade—a modified elemental rehydration solution composed of 9 selected amino acids. Our studies have shown that Enterade, which is undergoing clinical trials in 2 locations, (1) increases electrolyte absorption; (2) improves the mucosal barrier mechanism; (3) decreases anion and fluid secretions; (4) decreases osmotic diarrhea; and (5) improves appetite by decreasing nausea and vomiting and controlling the symptoms of GI toxicity. Since the beneficial effects of Enterade are obtained by tackling root physiological alterations, such as active chloride secretion and increased mucosal permeability, Enterade will be beneficial in other disease conditions with similar alterations in epithelial functions (e.g., inflammatory bowel diseases and infectious diarrhea).
Enterade may be beneficial for the following conditions: Diseases marked by villus atrophy with secondary malabsorption, increased paracellular permeability, and nausea and vomiting: Most absorptive processes are located in the fully differentiated and mature villous tip, with villous atrophy occurring secondary to reduced regeneration of intestinal epithelia following irradiation; these transport machineries are affected, which leads to malabsorption. The presence of unabsorbed or poorly absorbed nutrients and electrolytes alone can produce osmotic diarrhea. With increased epithelial permeability, there is increased bacterial translocation and susceptibility to infections. In addition, radiation exposure results in reduced bicarbonate secretions, which may cause acid peptic-like conditions that contribute to the increased nausea and vomiting associated with irradiation. Diseases associated with calcium-activated chloride secretion: Glucose stimulates active electrogenic sodium absorption via SGLT1 in the mammalian small intestine, a well-established physiological process that has been used to rehydrate patients with acute severe diarrhea. Although glucose enhances intestinal sodium and water absorption, which is the physiological basis for the development of oral

	rehydration solution (ORS) for acute diarrhea, this treatment is not associated with a dramatic reduction in stool output. The studies that led to the development of Enterade demonstrated that glucose also enhanced chloride secretion, which is mediated by an increase in intracellular calcium. Studies using gamma or proton irradiated mice demonstrated that both forms of radiation stimulated active chloride secretion via intracellular calcium. Since glucose also increases chloride secretion via increased intracellular calcium, glucose-based supportive therapy may not be the best agent for rehydration. Proton radiation-induced enteropathy: In addition to calcium-activated chloride secretion, proton radiation results in the activation of cAMP-mediated chloride secretion. Unlike the Ca-dependent secretagogues that often develop into tachyphylaxis, the cAMP-dependent secretagogues cause persistent secretion. Continued electrolyte loss in acute and chronic diarrhea can result in mineral loss that may indirectly affect bone mass and density.
Task Progress:	Over the past year, we examined the effects of proton- and gamma irradiation (0-1.5 Gy) on intestinal anion secretion and chloride secretion. We identified differing mechanisms of chloride transport for the two types of irradiation, and measured effects of gamma- and proton irradiation on mineral loss, focusing on stool loss of zinc, phosphorus, magnesium, and copper. Our further exploration of the effects of Enterade on irradiated and non-irradiated mice demonstrated that this selective amino acid-based rehydration solution has exciting potential with regards to improving mineral retention, electrolyte absorption, and intercellular permeability following radiation exposure.
Bibliography Type:	Description: (Last Updated: 08/24/2020)
Articles in Peer-reviewed Journals	Yin L, Gupta R, Vaught L, Grosche A, Okunieff P, Vidyasagar S. "An amino acid-based oral rehydration solution (AA-ORS) enhanced intestinal epithelial proliferation in mice exposed to radiation." Sci Rep. 2016 Nov 23;6:37220. https://doi.org/10.1038/srep37220 ; PMID: 27876791; PMCID: PMC5120277 , Nov-2016
Articles in Peer-reviewed Journals	Yin L, Menon R, Gupta R, Vaught L, Okunieff P, Vidyasagar S. "Glucose enhances rotavirus enterotoxin-induced intestinal chloride secretion." Pflugers Arch. 2017 Sep;469(9):1093-105. Epub 2017 May 10. https://doi.org/10.1007/s00424-017-1987-x ; PMID: 28488023 , Sep-2017
Articles in Peer-reviewed Journals	Yin L, Vaught L, Okunieff P, Casey-Sawicki K, Vidyasagar S. "Amino acid hydration decreases radiation-induced nausea in mice: A pica model." Adv Exp Med Biol. 2017;977:59-65. https://doi.org/10.1007/978-3-319-55231-6_9 ; PMID: 28685428 , Jul-2017